

Full length article

Band-structure reconstruction from solid-state high-harmonic generation via joint optimization of crystal potential and driving field

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ARTICLE INFO

Keywords:

High order harmonic generation
Attosecond pulse

ABSTRACT

Solid-state high-harmonic generation (HHG) offers a high-dimensional nonlinear observable that can be exploited for inverse reconstruction of material properties. In this work, we formulate the retrieval of solid-state structural information as a numerical inverse problem, in which the HHG spectrum serves as the optimization target. The forward model is based on solving the three-dimensional time-dependent Schrödinger equation (3D-TDSE) describing strong-field laser–solid interactions. A challenge in practical HHG-based inversion lies in the uncertainty of the driving infrared (IR) laser waveform, which can degrade reconstruction stability. To address this issue, we introduce a joint inversion strategy in which both the solid-state structural parameters and the temporal profile of the IR driving field are both treated as unknowns and are simultaneously optimized. This formulation reduces bias induced by imperfect laser characterization and improves the robustness of the inversion procedure. By combining first-principles strong-field simulations with data-driven optimization, the proposed framework enables stable and self-consistent reconstruction of material structural information from HHG signals. The method establishes a fully computational and all-optical approach for semiconductor characterization, with enhanced numerical stability against uncertainties in the driving field.

1. Introduction

Since its inception, solid-state HHG has emerged as a powerful tool with significant potential in both extreme light sources and all-optical schemes for probing material properties [1–11]. On the one hand, solid-state HHG provides a compact platform for generating high-order harmonics, offering a promising route toward tabletop extreme ultraviolet (XUV) and soft X-ray sources, and recent experiments have demonstrated the generation of sub-femtosecond VUV pulses with durations approaching the 950 as regime [12]. These extreme light sources are not only useful for ultrafast spectroscopy but also enable access to attosecond temporal resolution, opening new avenues for studying electron dynamics in condensed matter systems. On the other hand, solid-state HHG has demonstrated remarkable potential as an all-optical probe for characterizing semiconductor and solid-state structures [13–20]. Compared with traditional measurement techniques, such as angle-resolved photoemission spectroscopy (ARPES) or transport-based methods, HHG-based optical schemes are contact-free, can operate under strong-field conditions, and provide direct access to the electronic dynamics in the momentum space of the crystal.

In early studies, solid-state HHG was analyzed within the strong-field approximation (SFA) using incomplete saddle-point methods, which established a direct mapping between the HHG signal and band-structure-related quantities, such as the transition dipole moment between the valence and conduction bands [13–17]. While computationally efficient, these approaches rely on the restrictive assumption that laser-driven electronic transitions occur predominantly at a single crystal momentum point, typically the Γ point.

Subsequent works employed full numerical integration within the SFA framework to compute solid-state HHG spectra, thereby removing the single-point transition constraint [18]. However, these methods still assume negligible depletion of the valence band under strong laser excitation, which limits their applicability in more general regimes.

More recently, a fully numerical approach based on solving the three-dimensional SBE has been proposed to model solid-state HHG [21]. By applying inversion algorithms to the SBE formalism, the crystal potential can be reconstructed directly, enabling the retrieval of complete band-structure information. While this approach relaxes several key assumptions inherent in earlier schemes, it still assumes

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<https://doi.org/10.1016/j.commatsci.2026.114618>

Received 10 December 2025; Received in revised form 3 February 2026; Accepted 28 February 2026

Available online 18 March 2026

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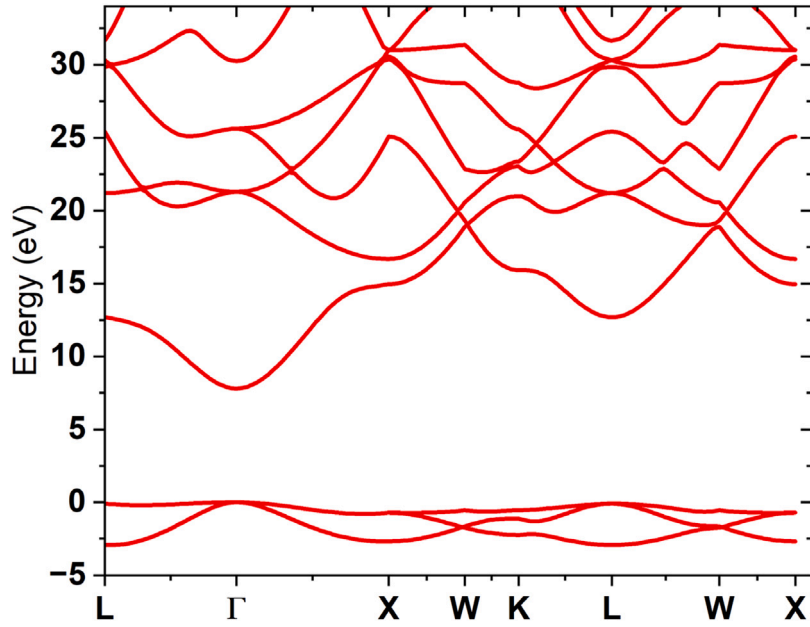


Fig. 1. Energy bandgap from our single electron simulation.

that the driving IR field is known. Numerical tests indicate that even with a 30% uncertainty in the IR field intensity, the reconstructed low-energy bands remain highly accurate, whereas noticeable deviations emerge in the higher-energy bands [21].

From an experimental perspective, the assumption that the driving IR field is precisely known can introduce systematic errors into HHG-based reconstruction schemes. In practice, the temporal waveform of the IR pulse is subject to uncertainties arising from pulse-to-pulse fluctuations and limited phase-resolved characterization accuracy [22]. These uncertainties are further amplified by strong-field nonlinearities, such that small deviations in the driving field can lead to significant changes in the generated HHG spectrum. When the IR field is treated as a fixed input in the inversion process, any mismatch is unavoidably transferred into the reconstructed material parameters, potentially biasing the retrieved band structure or crystal potential.

These considerations motivate a reformulation of the HHG-based inversion problem, in which the IR driving field is not assumed to be known *a priori*, but is treated as an additional unknown in the inversion procedure. The joint reconstruction framework employed in this work is designed to reduce error propagation from driving-field uncertainties and to enhance the numerical stability of HHG-based structural reconstruction.

In this work, we numerically solve the 3D-TDSE in momentum space under a periodic crystal potential to model strong laser–solid interactions. The resulting HHG spectrum serves as the forward model in an inverse reconstruction framework. In contrast to conventional approaches, both the crystal momentum-space potential $V(k_x, k_y, k_z)$ and the driving IR field characteristics, including the pulse envelope and carrier-envelope phase (CEP), are treated as unknown functions and are reconstructed simultaneously.

The inversion procedure employs a hybrid optimization strategy combining particle swarm optimization (PSO) and genetic algorithms (GA) to efficiently explore the high-dimensional parameter space. By explicitly incorporating the IR driving field into the reconstruction process, the proposed framework effectively reduces the impact of laser waveform uncertainties on reconstruction accuracy, leading to improved robustness in the retrieval of solid-state structural information.

Atomic units are used throughout unless otherwise stated.

2. Theory and method

Within the independent-particle approximation, the electronic structure of a crystalline solid is described by a single-electron Hamiltonian consisting of the kinetic energy operator and a periodic crystal potential $V(\mathbf{r})$. In reciprocal space, the electronic wave function is expanded using a plane-wave basis, leading to the following eigenvalue problem for each crystal momentum $\mathbf{k}$,

$$\sum_{\mathbf{G}'} \left[\frac{1}{2} |\mathbf{k} + \mathbf{G}|^2 \delta_{\mathbf{G}, \mathbf{G}'} + V(\mathbf{G} - \mathbf{G}') \right] C_{\mathbf{G}'} = E_{\mathbf{k}} C_{\mathbf{G}}, \quad (1)$$

where $\mathbf{G}$ and $\mathbf{G}'$ are reciprocal lattice vectors, $C_{\mathbf{G}}$ denote plane-wave expansion coefficients, and $E_{\mathbf{k}}$ represent the electronic band energies.

The reciprocal-space crystal potential is expressed as a sum over atomic contributions,

$$V(\mathbf{q}) = \sum_{\alpha} S_{\alpha}(\mathbf{q}) v_{\alpha}(\mathbf{q}), \quad (2)$$

where $\mathbf{q} = \mathbf{G} - \mathbf{G}'$, $S_{\alpha}(\mathbf{q}) = e^{-i\mathbf{q} \cdot \mathbf{r}_{\alpha}}$ is the structure factor determined by the atomic position $\mathbf{r}_{\alpha}$ in the unit cell, and $v_{\alpha}(\mathbf{q})$ denotes the atomic form factor of species α . This formulation is general and applies to arbitrary crystalline materials with periodic lattice symmetry.

To enable efficient and flexible band-structure reconstruction, the atomic form factors $v_{\alpha}(\mathbf{q})$ are represented using parametric models.

Here, we adopt a generalized multi-Gaussian representation of the atomic form factors,

$$v_{\alpha}(\mathbf{q}) = \sum_i v_{\alpha}^{(i)}(\mathbf{q}), \quad (3)$$

where each component takes the form

$$v_{\alpha}^{(i)}(\mathbf{q}) = -\frac{A_{ai}}{\Omega_{cell}} \left(\frac{\pi}{B_{ai}} \right)^{3/2} e^{-\frac{q^2}{4B_{ai}}}. \quad (4)$$

By adjusting the amplitudes A_{ai} and widths B_{ai} , different energy and length scales associated with specific orbital characters can be selectively controlled. This core-shell-like decomposition provides a physically interpretable and extensible parameterization applicable to a broad class of crystalline materials.

The complete set of adjustable parameters is denoted as

$$\mathbf{P} = \{A_{ai}, B_{ai}\}, \quad (5)$$

which fully determines the reciprocal-space crystal potential and the resulting electronic band structure. The construction of the three-dimensional stationary wavefunction with Bloch theorem follows Eqs. 1–13 in Section 2 of [23].

In Fig. 1, we get the MgO band structure calculated within our single-electron Schrödinger equation framework. In this example, we use 6 $\mathbf{P}$ parameters ($A_1 = 0.01$, $A_2 = 1.00$, $A_3 = 4.16406$, $B_1 = 1.00$, $B_2 = 2.00$ and $B_3 = 0.30$) to fit the bandgap. As shown, the band structure computed using our method captures the essential features of the energy bands. In particular, the energy differences at high-symmetry points are in general agreement with the Density Function Theory results in Fig. 4 of [24].

To account for uncertainties in the experimentally realized driving field, the IR laser pulse is explicitly incorporated into the reconstruction framework using a parametric representation. The time-dependent electric field is written as

$$E(t) = f(t) \cos(\omega t + \phi), \quad (6)$$

where ω denotes the carrier frequency, assumed to be known, ϕ is the carrier-envelope phase (CEP), and $f(t)$ represents the pulse envelope.

The envelope function $f(t)$ is expanded using a set of B-spline basis functions (See Section 2 in [22] for very detail),

$$f(t) = \sum_{i=1}^{N_b} a_i B_i(t), \quad (7)$$

where $B_i(t)$ are B-spline basis functions defined over a fixed temporal support, a_i are the corresponding expansion coefficients, and N_b denotes the number of spline basis functions.

The B-spline representation provides a flexible yet numerically stable description of smooth pulse envelopes while avoiding the need to assume a specific analytical pulse shape. In our simulation throughout this work, we employ 7th order B-Spline function with $N_b = 7$.

Within this formulation, the carrier frequency ω is fixed, whereas the CEP ϕ and the spline coefficients $\{a_i\}$ are treated as unknown parameters to be reconstructed. Together with the crystal potential parameters $\mathbf{P} = \{A_{ai}, B_{ai}\}$, these quantities are optimized simultaneously in the inverse reconstruction procedure.

The interaction between a crystalline solid and a strong IR laser field is modeled by numerically solving the 3D-TDSE in reciprocal space under the periodic potential $V(\mathbf{k})$. The numerical solution of the 3D-TDSE follows the approach presented in [23] (See Eqs. 15–21 therein for details) with our open source GPU-Matrix TDSE solver 3D-GTDSE [25], to which the reader is referred for implementation details.

For a given set of crystal potential parameters and driving-field characteristics, the TDSE simulation yields the time-dependent electron current or polarization, from which the solid-state HHG spectrum is obtained via Fourier transformation [23].

Within the present framework, the HHG spectrum serves as the forward observable in an inverse reconstruction problem. The fitness function erf is constructed as the sum of absolute deviations between the calculated and target HHG spectra within the harmonic order range from m to l :

$$erf(\mathbf{P}, a_i, \phi) = \sum_{N=l}^m \left| \{H_{\text{exp}}(N\omega) - H_{\text{guess}}(N\omega)\} \right|, \quad (8)$$

Because the intensities of low-order harmonics are significantly higher than those in the plateau region, their inclusion can dominate the optimization process and reduce reconstruction stability. Therefore, only plateau-region harmonics are used in the fitness evaluation.

The objective is to determine the parameter set $\mathbf{P}$, $\{a_i\}$ and ϕ that minimizes the deviation between the simulated HHG spectrum and a target spectrum. Due to the strong nonlinearity of laser-driven dynamics and the high dimensionality of the parameter space, the inverse problem is generally ill-conditioned and not amenable to local, gradient-based optimization techniques.

To achieve stable and efficient reconstruction, a hybrid global optimization strategy combining particle swarm optimization (PSO) and genetic algorithms (GA) is employed (See Appendix in [26] for detail). PSO is used to explore the parameter space and identify candidate solutions, while GA operations, including selection, crossover, and mutation, are applied to maintain population diversity and mitigate premature convergence.

At each iteration, a candidate parameter set defines a unique crystal potential, from which the band structure and corresponding HHG spectrum are computed using the forward TDSE solver. The fitness of each candidate is evaluated based on the spectral discrepancy, and the parameters are iteratively updated until convergence is achieved. This hybrid PSO–GA framework provides a general and robust approach for reconstructing electronic band structures from nonlinear optical observables and has proven effective in efficiently locating the vicinity of the global minimum in multi-parameter optimization tasks [18,22,26].

The flowchart of the reconstruction calculation is totally the same as that in [26].

The present reconstruction framework is based on several well-defined physical and numerical assumptions that determine its range of applicability. First, the electron dynamics are described within the independent-particle approximation by solving the TDSE for an effective single-electron Hamiltonian. Many-body effects, such as explicit electron–electron correlations, dynamical screening, and excitonic interactions, are not treated explicitly, but are instead assumed to be implicitly incorporated into the effective crystal potential. This approximation is expected to be valid for materials and excitation regimes where strong-field dynamics are dominated by band dispersion and single-particle transitions.

Second, the crystal potential is parameterized using a finite set of adjustable form factors in reciprocal space. While this representation provides a flexible and physically interpretable description of band structures, it may not capture extremely fine features arising from strongly correlated or highly localized electronic states (See Fig. 1). The reconstruction accuracy therefore depends on the adequacy of the chosen parameterization and the spectral information content of the HHG signal.

Third, the present formulation assumes coherent laser-driven dynamics and neglects dissipative processes such as electron–phonon scattering and dephasing. These effects are expected to become important for longer pulse durations or elevated lattice temperatures, and their inclusion would require extensions of the present framework to density-matrix or open-system descriptions.

Despite these limitations, the method is broadly applicable to a wide class of crystalline materials, including centrosymmetric and non-centrosymmetric solids, as well as low-dimensional systems. The framework is particularly well suited for wide-bandgap and moderately correlated materials under strong-field excitation, where solid-state HHG can be accurately described within a single-particle picture. Extensions to incorporate many-body effects, dissipation, and multi-electron dynamics represent promising directions for future work.

3. Results

In this section, we present representative numerical examples to demonstrate the performance and robustness of the proposed reconstruction framework. Two prototypical systems are considered: a three-dimensional non-centrosymmetric magnesium oxide (MgO) crystal, and a two-dimensional graphene lattice. For all cases, the driving laser electric field is polarized along the crystallographic [100] direction. This choice corresponds to a high-symmetry direction commonly adopted in solid-state HHG simulations, allowing a clear assessment of the reconstruction performance while avoiding additional anisotropy-induced complexity.

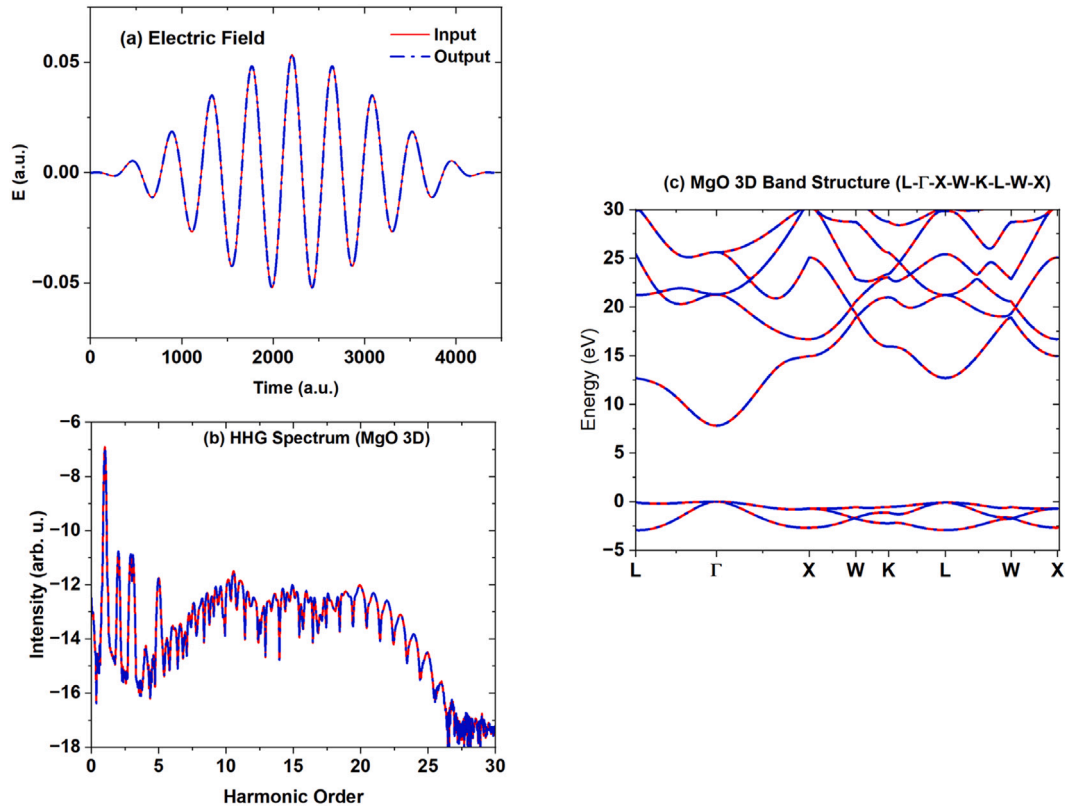


Fig. 2. Input and output electric field (a), HHG (b) and energy band (c). The red lines are the target ones while the blue dot lines are the retrieval data.

3.1. Non-centrosymmetric MgO crystal

We first consider MgO, which possesses a non-centrosymmetric symmetry. In this case, at least eight independent crystal potential parameters are required to capture the reduced symmetry. The reciprocal-space grid is increased to $101 \times 101 \times 101$ points to ensure adequate resolution. The driving IR laser pulse has a peak intensity of 1×10^{13} W/cm², a Gaussian envelope with a duration of 12 optical cycles, a central wavelength of 3200 nm, and a carrier-envelope phase (CEP) of $\phi = 0$. The reciprocal-space grid consists of $101 \times 101 \times 101$ k points, the time step is 0.05.

Convergence is determined by the fitness function (erf), defined as the fitness value of the optimization, reaching a threshold of 10^{-7} . Here, all HHG spectra are normalized such that the first harmonic has unit intensity. Using three NVIDIA RTX 3090 GPUs in parallel, the reconstruction converges after approximately 20 h of computation.

As shown in Fig. 2, the reconstructed band structure exhibits excellent agreement with the input band structure, with negligible deviations across the entire energy range. In addition, the reconstructed IR field waveform closely matches the input pulse. These results validate the accuracy of the proposed joint reconstruction algorithm for centrosymmetric three-dimensional solids.

Despite the enlarged parameter space and increased computational complexity, the algorithm successfully reconstructs both the crystal band structure and the IR driving field with high accuracy. The convergence is achieved after approximately 22 h of parallel computation. These results demonstrate that the proposed method remains effective for non-centrosymmetric materials, where symmetry-based parameter reduction is no longer applicable.

In solid-state HHG, the laser-driven electron dynamics are sensitive to both the temporal profile of the driving field and the spatial landscape of the crystal potential. By treating both as unknowns, our

framework exploits the distinct ways these factors encode information into the HHG spectrum. The driving field primarily dictates the sub-cycle timing and the maximum excursion amplitude of electrons in the Brillouin zone. In contrast, the crystal potential determines the band dispersion and the transition dipole moments. The simultaneous optimization allows the algorithm to decouple laser-induced phase shifts from band-structure-induced spectral features, preventing the “leakage” of experimental laser uncertainties into the reconstructed material parameters.

Unlike simplified models, the 3D-TDSE treats interband transitions and intraband acceleration on an equal footing. The low-order harmonics and the step-like features in the plateau are highly sensitive to the band curvature (effective mass) through intraband current, while the high-energy cutoff and specific peak intensities carry signatures of the bandgap and interband coupling. The joint inversion ensures that the energy gained by the electron from the field is physically consistent with the emitted photon energy, providing a self-consistent constraint that enhances the uniqueness of the solution.

3.2. Two-dimensional graphene

We next consider graphene as a representative two-dimensional material in Fig. 3. In this case, the simulation domain is reduced from three to two dimensions, leading to a substantial reduction in computational cost. The reciprocal-space grid is discretized using 101×101 k points, while the laser has a duration of about six optical cycles, 3×10^{12} W/cm² peak intensity and 3200 nm wavelength.

Owing to the reduced dimensionality, the reconstruction converges within approximately 2 h. The reconstructed band structure and IR field are in good agreement with the input results, confirming that the proposed framework can be efficiently extended to low-dimensional materials.

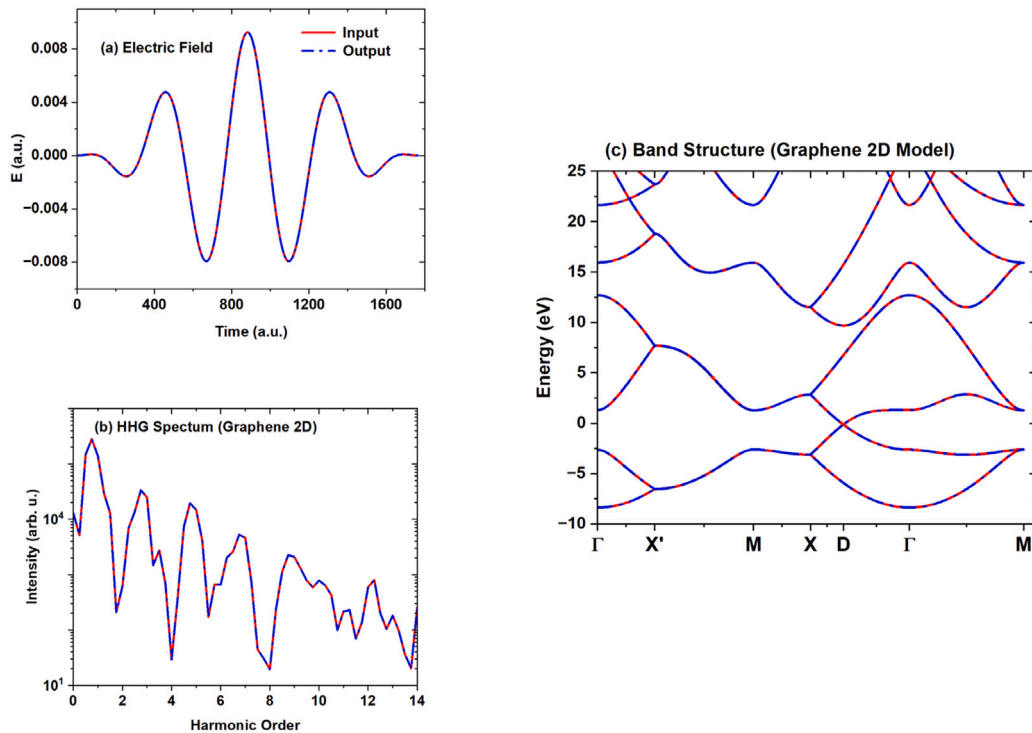


Fig. 3. The same calculations with Fig. 2, but the target is graphene.

3.3. Effect of noise

Finally, we investigate the robustness of the reconstruction against noise, which is unavoidable in experimental HHG measurements due to detection uncertainty and environmental fluctuations in Fig. 4. In this test, a random noise level of 10% is added independently to each harmonic order of the MgO HHG spectrum. The noisy spectrum is then used as the input for the reconstruction procedure. All the parameters are same as those in Fig. 2.

Taking the MgO crystal as an example, the results show that, despite the presence of significant noise, the essential features of both the electronic band structure and the IR driving field are successfully recovered. The convergence is reached after approximately 25 h of computation. This test demonstrates that the proposed method exhibits a high degree of tolerance to noise, highlighting its potential applicability to realistic experimental data.

The remarkable tolerance to 10% random noise can be attributed to the high dimensionality of the HHG observable relative to the compact parameter space of the B-spline and form-factor models. The TDSE acts as a “quantum-mechanical filter”: since random noise does not satisfy the underlying wave-equation dynamics, the optimization process naturally prioritizes spectral features that are physically reachable by the Hamiltonian. This inherent redundancy in the nonlinear signal ensures that the global topology of the fitness landscape remains dominated by the true physical parameters.

3.4. Laser polarization and crystal anisotropy

In the present work, the driving laser field is polarized along a fixed crystallographic direction, chosen as the high-symmetry [100] axis for three-dimensional crystals and an equivalent in-plane direction for two-dimensional materials. This choice is motivated by the goal of providing a clean and controlled benchmark for assessing the performance of the inverse reconstruction framework, while minimizing additional complexity arising from anisotropic strong-field responses.

It is important to note that the proposed method does not rely on any assumption of isotropic electronic structure or polarization-independent HHG emission. On the contrary, the HHG spectrum inherently encodes directional information through the crystal momentum dependence of the electronic band structure and transition matrix elements. The present formulation can be straightforwardly extended to arbitrary polarization directions or to multi-directional driving fields by modifying the coupling term in the TDSE. In such cases, the additional anisotropy is expected to provide complementary constraints that may further enhance the identifiability of the reconstructed band structure.

A systematic exploration of polarization-dependent HHG and its impact on reconstruction accuracy is beyond the scope of the present study and will be addressed in future work. Nevertheless, the results presented here demonstrate that even a single polarization configuration already contains sufficient information to enable robust reconstruction of both the crystal band structure and the driving field.

4. Conclusion

We have developed a general and fully computational framework for reconstructing electronic band structures of crystalline solids from solid-state high-harmonic generation (HHG) spectra. By numerically solving the three-dimensional time-dependent Schrödinger equation in reciprocal space, the HHG response is employed as a forward observable in a nonlinear inverse problem. Both the crystal potential and the driving infrared (IR) laser field are treated as unknowns and are reconstructed simultaneously, thereby mitigating systematic errors arising from uncertainties in laser characterization.

A flexible parametric representation of the crystal potential and the IR driving field is introduced, and a hybrid optimization strategy combining particle swarm optimization and genetic algorithms is employed to efficiently explore the high-dimensional parameter space. The method is validated using representative examples including non-centrosymmetric three-dimensional crystals, as well as a

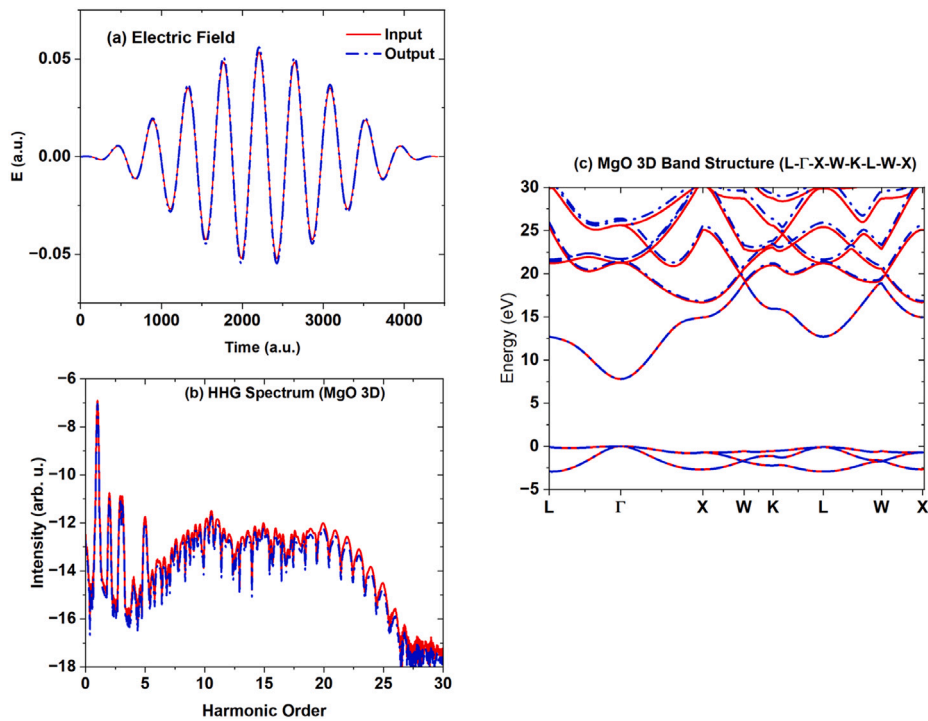


Fig. 4. Same calculation with that in Fig. 2, but with a noise of 10% percent.

two-dimensional material. The results demonstrate accurate reconstruction of both band structures and driving-field waveforms, with strong robustness against noise.

The proposed approach provides a complementary all-optical route for retrieving electronic structure information of solids and offers a versatile platform for future extensions, including polarization-dependent reconstruction, multi-band systems, and direct application to experimental HHG data.

CRediT authorship contribution statement

Lu Zheng: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Guoxiang Luo:** Methodology. **Jun Wang:** Methodology. **Xi Zhao:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Code availability

All codes used in this work are fully open, the TDSE solver can be downloaded from the Science Data Bank at <https://doi.org/10.57760/sciencedb.27107>. The optimization code is publicly available from the supplementary material of [26].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Guizhou Provincial Natural Science Foundation (Grant No. QianKeHeJiChu-MS [2026] 771).

Data availability

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

References

- [1] T. Schultz, M. Vrakking, *Attosecond and XUV Physics: Ultrafast Dynamics and Spectroscopy*, John Wiley & Sons, 2013.
- [2] S. Ghimire, A.D. DiChiara, E. Sistrunk, P. Agostini, L.F. DiMauro, D.A. Reis, *Nat. Phys.* 7 (2) (2011) 138–141.
- [3] E. Goulielmakis, T. Brabec, *Nat. Photonics* 16 (6) (2022) 411–421.
- [4] P. Wang, C.-Y. Luan, M. Qiao, M. Um, J. Zhang, Y. Wang, X. Yuan, M. Gu, J. Zhang, K. Kim, *Nat. Commun.* 12 (1) (2021) 233.
- [5] C. Lüders, M. Pukrop, F. Barkhausen, E. Rozas, C. Schneider, S. Höfling, J. Sperling, S. Schumacher, M. Aßmann, *Phys. Rev. Lett.* 130 (11) (2023) 113601.
- [6] M. Lostaglio, K. Korzekwa, D. Jennings, T. Rudolph, *Phys. Rev. X* 5 (2) (2015) 021001.
- [7] E. Bauch, C.A. Hart, J.M. Schloss, M.J. Turner, J.F. Barry, P. Kehayias, S. Singh, R.L. Walsworth, *Phys. Rev. X* 8 (3) (2018) 031025.
- [8] Y. Liu, B. Zhu, S. Jiang, S. Huang, M. Luo, S. Zhang, H. Yan, Y. Zhang, R. Lu, Z. Tao, *Phys. Rev. Lett.* 133 (2) (2024) 026901.
- [9] C. Qian, S. Jiang, T. Wu, H. Weng, C. Yu, R. Lu, *Phys. Rev. B* 109 (20) (2024) 205401.
- [10] T. Chang, T. Cohen, I. Holzman, G. Catelani, M. Stern, *Phys. Rev. Appl.* 19 (2) (2023) 024066.
- [11] H. Bartling, M. Abobeih, B. Pingault, M. Degen, S. Loenen, C. Bradley, J. Randall, M. Markham, D. Twitchen, T. Taminiau, *Phys. Rev. X* 12 (1) (2022) 011048.
- [12] A. Nayak, D. Rajak, B. Farkas, C. Granados, P. Stammer, J. Rivera-Dean, T. Lamprou, K. Varju, Y. Mairesse, M.F. Ciappina, M. Lewenstein, P. Tzallas, *Nat. Commun.* 16 (1) (2025) 1428.
- [13] G. Vampa, T. Hammond, N. Thiré, B. Schmidt, F. Légaré, C. McDonald, T. Brabec, D. Klug, P. Corkum, *Phys. Rev. Lett.* 115 (19) (2015) 193603.
- [14] Y.-T. Zhao, S.-y. Ma, S.-C. Jiang, Y.-J. Yang, X. Zhao, J.-G. Chen, *Opt. Express* 27 (23) (2019) 34392–34404.
- [15] Y. Qiao, Y. Huo, H. Liang, J. Chen, W. Liu, Y. Yang, S. Jiang, *Phys. Rev. B* 107 (7) (2023) 075201.
- [16] D. Wu, L. Li, Y. Zhan, T. Huang, H. Cui, J. Li, P. Lan, P. Lu, *Phys. Rev. A* 105 (6) (2022) 063101.
- [17] K. Uchida, V. Pareek, K. Nagai, K. Dani, K. Tanaka, *Phys. Rev. B* 103 (16) (2021) L161406.
- [18] J. Zhang, J. Wang, X. Zhao, *Phys. Rev. A* 112 (2025) 043103.

- [19] J. Zhang, Z. Wang, F. Lengers, D. Wigger, D.E. Reiter, T. Kuhn, H.J. Wörner, T.T. Luu, *Nat. Photonics* 18 (8) (2024) 792–798.
- [20] W. Li, X. Zhu, P. Lan, L. Li, K. Wang, W. He, H. Hübener, U. De Giovannini, P. Lu, *Phys. Rev. Lett.* 135 (2025) 086404.
- [21] A.M. Parks, M. Kolesik, *Opt. Express* 33 (6) (2025) 13986–13999.
- [22] X. Zhao, H. Wei, Y. Wu, C.D. Lin, *Phys. Rev. A* 95 (2017) 043407.
- [23] X.-Q. Wang, X.-B. Bian, *Phys. Rev. A* 103 (2021) 053106.
- [24] Y. Tan, M. Povolotskyi, T. Kubis, Y. He, Z. Jiang, G. Klimeck, T.B. Boykin, J. Comput. Electron. 12 (1) (2013) 56–60.
- [25] K. Peng, A. Liu, J. Wang, X. Zhao, *Chin. Phys. B* 34 (9) (2025) 094203.
- [26] J. Liu, S. Jiang, J. Chen, L. He, J. Wang, X. Zhao, *APL Comput. Phys.* 1 (2) (2025) 026104.